Project Title

Socioeconomic and Demographic Drivers of Electric Vehicle Adoption in Washington State

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1. Research Question

How do median household income, population size, and median age at the ZIP-code level influence the rate of electric vehicle adoption (BEVs and PHEVs) across Washington State?

2. Project Objectives

The objectives of this project are to quantify electric vehicle adoption rates at the ZIP-code level across Washington State and to analyze the statistical relationships between EV adoption density and median income, population size, and median age. The study aims to identify socioeconomic and demographic disparities in electric vehicle adoption and to evaluate which variables are most strongly associated with higher levels of EV penetration. All analyses will be conducted using reproducible methods and will follow responsible data practices, with findings interpreted in the context of transportation equity and public policy.

3. Dataset Description

The primary dataset is the Electric Vehicle Population Data (2025) obtained from the Washington State Department of Licensing via Data.gov(https://catalog.data.gov/dataset/electric-vehicle-population-data).The dataset contains approximately 200,000 records of registered Battery Electric Vehicles (BEVs) and Plug-in Hybrid Electric Vehicles (PHEVs). Key variables include ZIP code, vehicle type, make, model year, electric range, and registration count. The dataset provides statewide coverage across Washington ZIP codes,

Supplementary data include population size and median age by ZIP code(2023), obtained from Data Commons using U.S. Census–derived indicators(https://datacommons.org/ranking/Median_Income_Household/).These data are accessed programmatically through the Data Commons API(https://apikeys.datacommons.org/my-apps/838c5d9d-8d45-432c-8e65-2f3205a78146).

All socioeconomic variables are aligned at the ZIP-code level and integrated with the electric vehicle dataset to support aggregated demographic and socioeconomic analysis.

4. Ethical and Data Concerns

- **Data Privacy:** All datasets are analyzed at the ZIP-code level and do not contain direct personally identifiable information (PII); however, the Vehicle Identification Number (VIN) will be masked to further reduce privacy risk.
- **Bias:** EV registration data may reflect charging infrastructure availability and affordability rather than purely consumer preference, which may lead to the underrepresentation of rural or lower-income communities.
- **Data Limitations:** ZIP-code–level aggregation may mask within-area socioeconomic inequality, and some ZIP codes may have sparse population or vehicle counts. Data accessibility and completeness across all ZIP codes in the 39 counties of Washington State have not yet been fully validated and will be documented as a limitation of the study.

5. Mitigation Measures

Electric vehicle counts will be normalized by population at the ZIP-code or city level to allow fair comparison across regions of different sizes. ZIP codes with missing data or low vehicle coverage will be identified and documented to maintain transparency. All analytical assumptions and data limitations will be clearly stated, and findings will be interpreted as statistical associations rather than causal relationships.

6. Success Metrics

Project success will be evaluated using correlation coefficients and regression model results that measure the strength and direction of relationships between EV adoption rates and socioeconomic variables. Model quality will be assessed through statistical significance, effect sizes, and coefficient interpretability. Visualizations such as income-versus-adoption plots and geographic summaries will be used to support insight-based evaluation, and all analysis steps will be documented to ensure reproducibility and transparency.